

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

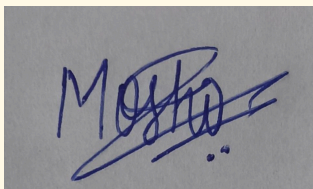
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech (AI-DS)	Date:	20.08.2026

Code of Conduct

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

ANSWER:

(a) Referring Expressions and Possible Antecedents

Referring Expression	Type	Possible Antecedents (before constraints)
John	Discourse-new entity (proper noun)	None (introduces entity)
Mary	Discourse-new entity (proper noun)	None (introduces entity)
the park	Discourse-new entity (location)	None (introduces entity)
He	Pronoun (masc., sing.)	John, Mary
a ball	Discourse-new entity (object)	None (introduces entity)
She	Pronoun (fem., sing.)	John, Mary
it	Pronoun (neuter, sing.)	the park, a ball
The dog	Discourse-new entity (animal)	None (introduces entity)
him	Pronoun (masc., sing.)	John, the dog
they	Pronoun (plural)	John, Mary, the dog (in combination)

(b) Applying the Resolution Constraints

Gender and number agreement: This constraint matches the morphological features of a pronoun to its candidates. "He" is masculine singular, so it agrees only with John and is incompatible with Mary (feminine) and the park (inanimate, no gender). "She" is feminine singular, so it agrees only with Mary. "They" is plural, so it can only refer to a group, eliminating any single-entity candidate.

Recency: This constraint prefers the antecedent that was mentioned most recently in the discourse, since working memory favors nearby entities. For "it", both "the park" and "a ball" are inanimate, but "a ball" was introduced more recently than "the park", so recency favors the ball.

Semantic compatibility (animacy and action suitability): This constraint checks whether the candidate's real-world properties fit the action described. "Play with it" requires a manipulable object, which fits "a ball" but not "the park" (a location is not something one plays with in this sense), confirming the recency-based choice. Similarly, "chased him" requires an animate patient capable of being chased; this fits John, but the subject of the same clause ("the dog") is syntactically barred from referring to itself as the object, so "him" cannot mean the dog.

Coherence (consistent entity chains): This constraint prefers interpretations that keep the discourse about the same set of entities consistently, rather than introducing unnecessary new referents. "They all went home" is most coherent when read as referring to every participant already active in the discourse: John, Mary, and the dog, since "all" signals inclusion of the whole established group.

(c) Constraint Table

Variable	Candidate Domain	Constraints Applied	Eliminated Candidates	Final Antecedent
He	{John, Mary}	Gender agreement	Mary (wrong gender)	John
a ball	{John, Mary, the ball}	Discourse-new (no antecedent needed)	-	New entity
She	{John, Mary}	Gender agreement	John (wrong gender)	Mary
it	{the park, a ball}	Recency + Semantic compatibility	the park (less recent, not playable)	a ball
The dog	{John, Mary, the dog}	Discourse-new (no antecedent needed)	-	New entity
him	{John, the dog}	Semantic compatibility + syntactic disjointness	the dog (subject of same clause)	John
they	{John, Mary, the dog, combinations}	Number agreement + Coherence	any single entity	John + Mary + the dog

Simple black-and-white constraint graph (variables as nodes, arrows to surviving candidate):

[He] -----> John

[She] -----> Mary

[it] -----> a ball (park eliminated)

[him] -----> John (dog eliminated: same-clause subject)

[they] -----> {John, Mary, the dog}

(d) Final Coreference Chains and Rewritten Paragraph

Chain	Members
Chain 1	John, He, him
Chain 2	Mary, She
Chain 3	the park
Chain 4	a ball, it
Chain 5	the dog
Chain 6 (group)	they = {John, Mary, the dog}

Rewritten paragraph: "John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home."

(e) Priority Order of Constraints and Effect of Relaxing Recency

Priority order (highest to lowest): (1) Gender and number agreement, (2) Semantic compatibility (animacy and action suitability), (3) Coherence, (4) Recency. Agreement is placed first because it is a hard, categorical constraint: a violation makes an antecedent grammatically impossible, not merely unlikely. Semantic compatibility is next because even grammatically valid candidates can be nonsensical in context (a park cannot be "played with"). Coherence is placed above recency because keeping a consistent entity chain across the whole discourse is a stronger cue to correct interpretation than closeness alone. Recency is used last, as a soft tie-breaker only when multiple candidates survive the stronger constraints.

If the recency constraint is relaxed, "it" becomes ambiguous between "the park" and "a ball", since both are inanimate and grammatically valid. In this example the semantic compatibility constraint (an object one can "play with") still favors the ball, so the correct resolution can be recovered from world knowledge; however, in general, relaxing recency increases ambiguity and forces the resolver to depend more heavily on deeper semantic or pragmatic reasoning, which is computationally more expensive and less reliable than a simple recency heuristic.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

ANSWER:

(a) Discourse Planning Steps Before Generation

1. Goal identification: fix the communicative goal as a combination of two dialog acts, Advise and Encourage.
2. Entity extraction: identify the key discourse entities from the user turn - "exam", "concentrate", and "you" (the user) - to be reused for coherence.
3. Content selection: choose an actionable piece of advice (for example, taking a short break) and a supportive statement expressing confidence in the user.
4. Discourse relation selection: decide how the advice and encouragement clauses will be connected, using Cause-Effect (lack of concentration leads to the suggested action) and Elaboration (the encouraging clause adds supportive detail).
5. Lexical selection: ensure at least two of the required keywords (focus, break, confident) are included.
6. Surface realization: generate a grammatical, polite, positive-toned response of exactly 2-3 sentences.
7. Constraint verification: check the generated text against all six constraints before finalizing it.

(b) Three Candidate Responses

R1: "Take a short break before you go back to your exam preparation - a few minutes away will help you concentrate better. You have prepared well, so stay confident going into the exam."

R2: "Since you are finding it hard to concentrate on your exam, try taking a ten-minute break to reset your mind. Afterward you will likely regain your focus, and you should feel confident that your preparation will carry you through."

R3: "It is understandable that an important exam makes it harder for you to concentrate, so take a short break to relax before continuing your revision. Once you resume, aim to bring steady focus, and remind yourself that you are confident and capable of doing well."

(c) Evaluation of the Three Responses

Criterion	R1	R2	R3
Advice present	Yes	Yes	Yes
Encouragement present	Yes	Yes	Yes
Entity coherence (exam, concentrate, you)	Yes	Yes	Yes
Discourse relation (Cause-Effect / Elaboration)	Cause-Effect	Cause-Effect + Elaboration	Cause-Effect + Elaboration
Required keywords (≥ 2 of focus/break/confident)	break, confident	break, focus, confident	break, focus, confident
Sentence count (2-3)	2	2	2
Logical consistency	Yes	Yes	Yes
Positive, polite tone	Yes	Yes	Yes

Best response: R2. Justification - R2 satisfies every constraint at least as well as R1 and R3, and it is the most tightly coherent: it explicitly states the cause ("finding it hard to concentrate on your exam") before proposing the effect ("try taking a ten-minute break"), giving a clean Cause-Effect structure, and then adds an Elaboration clause that links the break to regaining focus and feeling confident. All three required keywords appear naturally rather than being inserted mechanically, and the entity chain (exam, concentrate, you) is reused smoothly across both sentences, giving it the strongest discourse coherence of the three.

(d) Effect of Violating Any Two Constraints

If, for example, the keyword constraint and the sentence-length constraint were both violated (say, only one required keyword is used and the response runs to five sentences), the output would drift from the intended Advise + Encourage act: the excessive length would make the response feel like a lecture rather than a concise supportive reply, weakening the Encourage component, while the missing keywords would make the advice feel vague and generic. More generally, violating two constraints compounds their individual effects: the response may still be grammatical, but it becomes less targeted, less coherent with the required dialog act, and less natural in tone, reducing the user's trust that the system understood their specific situation.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

ANSWER:

(a) Resolving the Ambiguous Word "bank"

Resolved sense: riverbank (the land alongside a river), not the financial institution.

Justification using the given constraints: (1) Correct WSD using context - the phrase "by the river" is a strong local collocation for the landform sense, since a financial institution is not normally described as being spatially "by the river" in this way. (2) Predicate compatibility - the predicate flood(x) selects for a low-lying natural location as its argument; a riverbank satisfies this selectional restriction naturally, while a financial institution would only "flood" in an unusual, metaphorical, or building-damage sense, which is not indicated anywhere else in the sentence. (3) Discourse and coherence constraints - the sentence continues with "it was saved by quick action", which coheres with an image of a riverbank being reinforced or protected (for example with sandbags) during a storm, a scenario much more typical and coherent than a bank building being "saved". Together these constraints eliminate the financial-institution sense and confirm the riverbank.

(b) Predicate Logic Representation

VARIABLES:

b = the riverbank

r = the river

s = the storm

e1 = the flooding event

e2 = the saving event

a = the quick action

PREDICATES:

bank(b) - b is a riverbank

river(r) - r is a river

storm(s) - s is a storm

locatedNear(b,r) - the bank is located by the river

flood(e1) - e1 is a flooding event

theme(e1,b) - b is the affected entity of e1

causedBy(e1,s) - the flooding was caused by the storm

save(e2) - e2 is a saving event

theme(e2,b) - b is the affected entity of e2

action(a) - a is an action

quick(a) - a is quick

instrument(e2,a) - a is the instrument/means of e2

after(e1,s) - e1 occurred after s

FINAL REPRESENTATION:

Exists b,r,s,e1,e2,a [

bank(b) AND river(r) AND locatedNear(b,r) AND storm(s) AND

flood(e1) AND theme(e1,b) AND causedBy(e1,s) AND after(e1,s) AND

save(e2) AND theme(e2,b) AND action(a) AND quick(a) AND

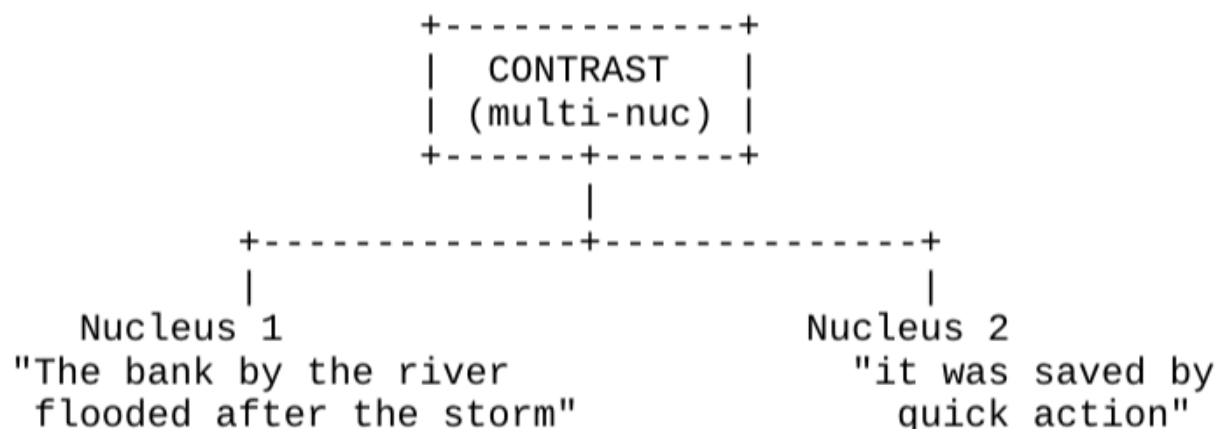
instrument(e2,a)

]

(c) Target-Language Sentence (Meaning-Preserving Paraphrase)

"The riverbank flooded because of the storm, but quick action saved it." This paraphrase preserves all entities (riverbank, storm, saving action), the causal relation between the storm and the flood, and the Contrast relation between the flooding and the successful rescue, while removing the ambiguity of the word "bank" by making the riverbank sense explicit.

(d) RST-Style Discourse Tree Showing Contrast



Both spans are treated as nuclei of a multi-nuclear Contrast relation because neither clause is subordinate in importance: the first presents an adverse event (flooding) and the second presents an opposing, favorable outcome (being saved), and both are essential to the intended meaning.

(e) Constraint-Based Approach vs Pure Statistical MT

Aspect	Constraint-Based Approach	Pure Statistical MT
Ambiguity resolution	Uses explicit selectional and contextual constraints to resolve "bank" correctly and deterministically	Relies on statistical co-occurrence patterns; can pick the more frequent sense (financial institution) regardless of local context
Semantic consistency	Predicate logic enforces that arguments match predicate requirements (e.g. flood(x) needs a location)	No explicit semantic checking; output may be locally fluent but semantically inconsistent
Discourse preservation	Explicitly represents and preserves the Contrast relation between clauses	May preserve surface connectives like "but" without guaranteeing the underlying discourse relation is maintained in meaning
Entity/reference consistency	Entities (bank, river, storm, action) are tracked explicitly across the representation	Entity tracking is implicit and can break down across longer or more complex sentences
Interpretability	Every decision (sense choice, predicate structure) is traceable to an explicit rule or constraint	Decisions emerge from learned weights and are largely opaque
Information preservation	Guarantees preservation of entities, roles, and relations because they are represented explicitly	May drop, add, or distort information, particularly for ambiguous or low-frequency constructions

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____